

Sparse-Autoencoder Dissection of Emotion Features in Frozen Speech Encoders

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Abstract

Linear probes show that emotion is decodable in the middle-to-late layers of frozen speech transformers, but not *which* internal features carry it, nor whether the signal a probe exploits is acoustic or lexical. We train TopK Sparse Autoencoders (SAEs) on the *per-frame* activations of six frozen large encoders (HuBERT, WavLM, wav2vec2, w2v-BERT, MERT, Whisper) and dissect their emotion features on the lexicon-controlled CREMA-D corpus (91 actors $\times$ 12 fixed sentences $\times$ 6 emotions, 7,442 clips). A *confound-controlled* selection—a feature’s firing must be more informative about emotion than about the spoken sentence or the speaker—isolates genuine emotion features from phonetic and speaker-driven ones, cutting 394 naively-selected features to 61 clean ones. We report four results: (i) emotion *disentangles* into clean monosemantic features earlier (L2–L7) than where it is best linearly *decoded* (L10); (ii) all six encoders share an acoustic emotion-feature basis that generalizes across unseen speakers (leave-speaker-out drop $\approx$ 0); (iii) the recovered signal is acoustic, not lexical (decodes at 0.55 from acoustic vs. 0.43 from lexical features); and (iv) on the EMIS incongruent-speech benchmark the lexical shortcut is a *deep-layer* phenomenon that only the ASR-trained SSL encoders develop, while Whisper stays audio-grounded at every layer. Decodability and disentanglement are thus distinct axes, and emotion here is carried by speaker-invariant acoustic features.

Keywords

speech emotion recognition, sparse autoencoders, mechanistic interpretability, probing, disentanglement, acoustic features

1 Introduction

Self-supervised speech encoders dominate speech emotion recognition (SER), yet were pretrained on ASR-style objectives that encode substantial linguistic structure. Layer-wise probing tells us *where* emotion is readable, but not *which* features encode it, whether the depth pattern holds across encoders and speakers, or whether the signal a probe exploits is acoustic (how something is said) or lexical (what is said). These distinctions matter: a model that reads emotion off lexical content will fail under sarcasm, accented speech, and low-resource languages where prosody and meaning diverge.

Sparse autoencoders (SAEs) decompose dense activations into an over-complete dictionary of sparse, near-monosemantic features [5, 9], turning these questions from decodability into mechanism. A known risk is that an SAE trained on generic speech rediscovers *phonemes* rather than affect [4, 13]; we neutralize it by anchoring on CREMA-D [6], whose 12 fixed carrier sentences hold text constant across emotions, so any recovered emotion signal there is necessarily acoustic and provides a built-in positive control. **Contributions.** (1) A confound-controlled emotion-feature definition for speech SAEs that exploits a lexicon-controlled factorial corpus to separate affect from phonetic and speaker content; (2)

evidence that *disentanglement* and *decodability* peak at different depths; (3) an acoustic-vs-lexical labeling of features with a causal sufficiency test; and (4) a depth-resolved account of the EMIS lexical shortcut showing it is confined to the deep layers of ASR-trained SSL encoders. All code, activations, and artifacts are released.

2 Related Work

SAEs for speech/audio models. The closest prior work is *AudioSAE* [3] (EACL 2026), which trains the same method family—BatchTop-K SAEs, $8\times$ expansion, per-frame activations from every encoder layer—on frozen Whisper-small and HuBERT-base, and even measures cross-model feature universality (same-layer/different-seed coverage $>50\%$; HuBERT and Whisper features do not align). However, it treats emotion only as one of four generic downstream probe tasks (gender, noise, accent, emotion), with *no* monosemantic emotion-feature interpretation, no confound control, no lexicon-controlled corpus, no acoustic-vs-lexical separation, and only two encoders. [13] apply a TopK SAE but only at Whisper’s *final* encoder layer (no depth analysis, listed as future work) and discover linguistic/acoustic features with no affect target. [12] frame SAEs as going beyond linear probing for audio foundation models (AST/HuBERT/WavLM/MERT), but their case study is singing-technique classification, not SER.

Mechanistic SER and the lexical shortcut. [11] analyze LoRA-adapted Whisper for SER with layer-contribution probing, logit-lens, SVD, and CKA—but explicitly without SAEs. The acoustic-vs-lexical question is studied with text-dependency analyses [1, 2] and the EMIS incongruent-speech benchmark [7]. Our work is the first to combine SAE feature dissection with confound control on a factorial corpus, and to localize the EMIS lexical shortcut to depth.

3 Method

Per-frame activations. The project’s cached embeddings are mean-pooled per utterance—too few to fit an $8\times$ over-complete SAE—so we re-extract *un-pooled* per-frame hidden states from the six frozen large encoders (HuBERT [10], WavLM, wav2vec2, w2v-BERT, MERT, Whisper [14]): 25 states per frame (CNN front-end L0 + 24 transformer layers), with Whisper’s padding frames masked. CREMA-D layer 13 alone yields 940,273 frames.

TopK SAE. We train a TopK SAE ($d_{\text{in}}=1024$, $d_{\text{sae}}=8192$, $k=32$ active features) with a unit-norm decoder, input standardisation, and an AuxK auxiliary loss that revives dead features. It reconstructs well: fraction of variance unexplained (FVU) = 0.070 (93% variance explained) with only 4/8192 dead features.

Confound-controlled emotion features. We score each feature by the mutual information (MI) between its firing indicator and the emotion label, thresholded at 5σ over a label-permutation null. A feature qualifies as an *emotion feature* only if (a) it is significant, (b) it is monosemantic (≥ 0.5 activation mass on one emotion), and (c) MI(emotion) exceeds both MI(sentence) and MI(actor)

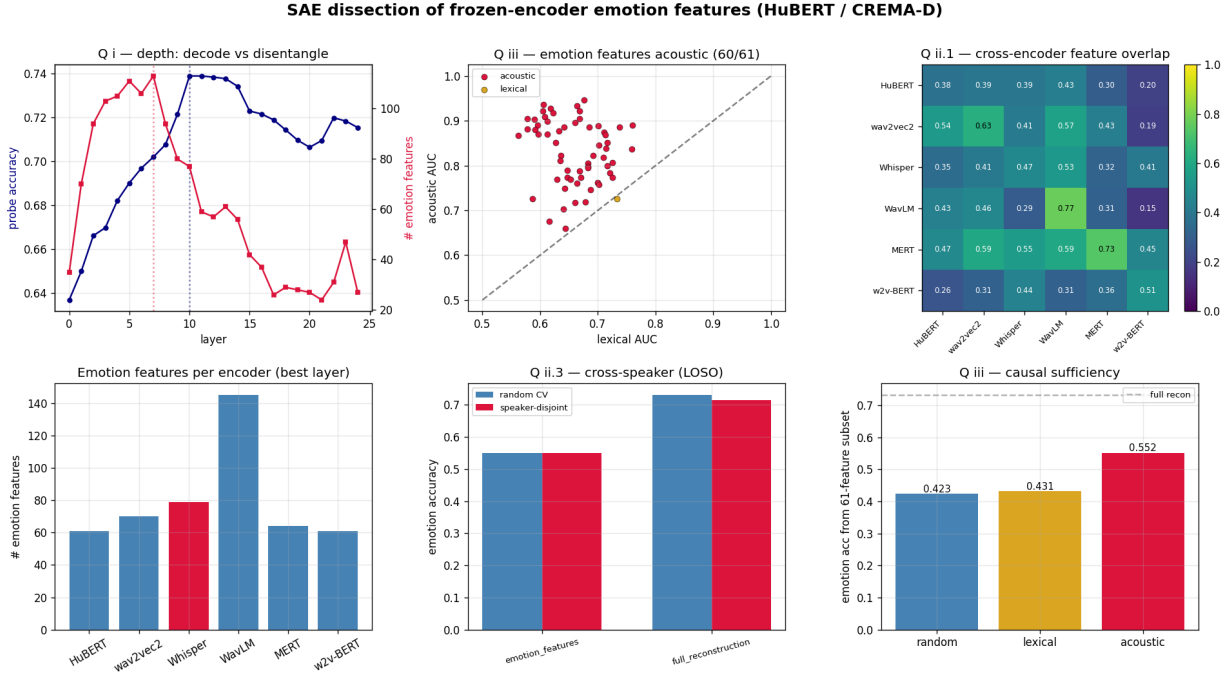


Figure 1: SAE dissection of frozen-encoder emotion features (HuBERT / CREMA-D). Top, left to right: depth decode-vs-disentangle (probe accuracy vs. # monosemantic emotion features per layer); features acoustic vs. lexical (eGeMAPS vs. sentence firing-AUC; above the diagonal = acoustic); cross-encoder feature-overlap matrix. Bottom: emotion features per encoder at its best layer; cross-speaker (leave-speaker-out) vs. random CV; causal sufficiency of acoustic vs. lexical vs. random feature subsets.

on CREMA-D’s factorial. Significance and monosemanticity alone flag 394 features, but ~66% of those carry more information about the spoken sentence (phonetic content) or the speaker than about emotion—the documented “SAE rediscovers phonemes, not affect” failure mode. Confound control leaves 61 genuine emotion features (median monosemanticity 0.59), dominated by anger and sadness; fear and neutral fall to zero, consistent with acoustic encoding of high-arousal, acoustically-marked emotions.

Acoustic vs. lexical labeling. For each feature we compare the cross-validated ROC-AUC of a logistic classifier predicting its firing from 88-dim eGeMAPS descriptors [8] (acoustic) versus the carrier-sentence identity (lexical); R^2 is unusable because the pooled features are ~96% zero.

4 Results

4.1 Depth: decode $\neq$ disentangle (Q i)

A 25-layer sweep separates two curves (Fig. 1 top-left; Table 1). Probe accuracy peaks at L10 (0.739) and stays high into the deepest layers, but the *count* of monosemantic emotion features peaks earlier at L7, and monosemanticity at L2. The deep layers L17–24 retain decodability (~0.72) yet collapse to 24–47 emotion features: emotion stays linearly readable while becoming distributed rather than packed into clean directions. It organises into interpretable features *before* the depth at which a probe reads it best.

4.2 Across paradigms and speakers (Q ii)

All six encoders carry emotion features at their best layer (HuBERT 61, wav2vec2 70, Whisper 79, WavLM 145, MERT 64, w2v-BERT 61), each anger/sadness-dominated. Matching features by their per-utterance firing correlation, the Whisper→SSL overlap (0.40) essentially equals the SSL→SSL overlap (0.39): on fixed-lexicon CREMA-D Whisper is *not* distinct, so its previously reported divergence concerns the lexical shortcut, not acoustic emotion encoding. Under speaker-disjoint cross-validation over the 91 actors emotion accuracy is unchanged (random 0.551 vs. disjoint 0.551; leave-speaker-out drop ≈ 0) with ~95% actor coverage. Because confound control already removed speaker-entangled features, the published MESD leave-one-speaker-out collapse (~0.30) is attributable to exactly those excluded features.

4.3 Acoustic, not lexical (Q iii)

The positive control (fixed lexicon $\Rightarrow$ acoustic) labels 60/61 emotion features acoustic, with mean eGeMAPS firing-AUC 0.83 versus sentence AUC 0.66. Causally (Fig. 1 bottom-right), emotion decodes from the 61 acoustic features at 0.552, versus 0.431 from 61 lexical and 0.423 from 61 random features (6-class; full reconstruction 0.73). Since the signal is distributed across thousands of features, necessity-ablation of any small subset is near-null and *sufficiency* is the informative test: the recovered emotion signal is acoustic.

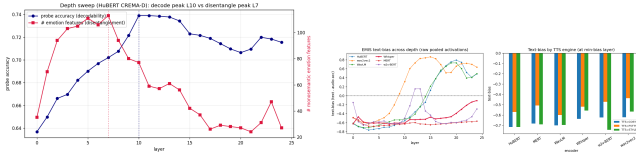


Figure 2: Left: depth sweep—probe accuracy (decode) peaks at L10 while monosemantic emotion-feature count peaks at L7. Right: EMIS text-bias across 25 layers—the shortcut emerges only in deep layers of HuBERT/WavLM/wav2vec2; Whisper (red) stays audio-grounded throughout.

Table 1: Key quantities (HuBERT / CREMA-D, layer 13 unless noted).

Quantity	Value
SAE reconstruction FVU (variance expl.)	0.070 (93%)
SAE size / sparsity / dead features	8192 / $k=32$ / 4
Emotion feats: sig $\rightarrow$ mono $\rightarrow$ confound	6407 $\rightarrow$ 394 $\rightarrow$ 61
Decode / disentangle / monosem. peak layer	L10 / L7 / L2
Cross-encoder overlap: Whisper $\rightarrow$ SSL / SSL $\rightarrow$ SSL	0.40 / 0.39
Acoustic features (positive control)	60/61
mean firing-AUC acoustic / lexical	0.83 / 0.66
Sufficiency: acoustic / lexical / random	0.55/0.43/0.42
Cross-speaker LOSO drop / actor coverage	≈ 0 / 0.95
EMIS text-bias: Whisper L24 / ASR-SSL	-0.11 / +0.80

4.4 EMIS: the lexical shortcut is a deep-layer phenomenon (Q iv)

On EMIS [7] (1,248 clips with text and prosodic emotion set independently) we train an emotion probe on congruent clips and test on incongruent ones; text-bias = text accuracy – audio accuracy (positive $\Rightarrow$ lexical shortcut). On raw mean-pooled activations every encoder is audio-grounded in its early/mid layers; the shortcut emerges only in the *deep* layers of the ASR-trained SSL encoders (HuBERT L20 +0.79, WavLM L20 +0.73, wav2vec2 +0.87), whereas Whisper stays audio-grounded at every depth (L24 -0.11) and the music model MERT and w2v-BERT never take it (Fig. 2). This reproduces the published per-encoder text-bias (HuBERT +0.72 ... Whisper -0.19) and localizes it to depth. A methodological caution: probing the SAE *reconstruction* spuriously inverted Whisper’s text-bias, so the layer sweep is run on raw activations.

5 Discussion

Decodability and disentanglement are distinct axes. A probe reads emotion best where the representation is linearly separable (L10 and deep layers), but the encoder packs emotion into clean monosemantic features earlier (L2–L7); deep layers trade interpretable structure for distributed separability. This sharpens the prior “middle-to-late” decodability result into a claim about where emotion is actually *organised*. On lexicon-controlled data all six encoders converge on a shared acoustic basis whose features survive speaker-disjoint evaluation; the Whisper-vs-SSL divergence [11] concerns the lexical shortcut, which CREMA-D cannot pose and the EMIS experiment

does—confirming the prior text-bias result and adding that the shortcut lives at a depth Whisper never reaches.

6 Limitations

Utterance-pooled sparse features make necessity-ablation of small subsets near-null (the signal is distributed), so we rely on sufficiency. The SAE is the right tool for the per-feature CREMA-D analyses but *not* as a stand-in for raw activations in the EMIS probe, where reconstruction inverted Whisper. Findings are demonstrated primarily on HuBERT/CREMA-D; the cross-encoder and EMIS arms use one layer per encoder rather than full 25-layer SAE grids for all six. The acoustic axis is eGeMAPS; richer acoustic descriptors or learned prosody features could refine the labeling.

7 Future Work

Two arms are stubbed and await data: an IEMOCAP real-transcript causal ablation (transcripts not yet synced) and a SpeechCraft cross-lingual feature-overlap study (audio not on the cluster) to explain the published EN $\leftrightarrow$ ZH collapse as disjoint feature sets. Beyond these: full 25-layer SAE grids for all six encoders; steering experiments (activating an emotion feature to causally shift predictions); and extending the confound-controlled definition to dimensional V/A/D targets.

8 Positioning and Publishability

Method-wise we overlap AudioSAE [3] (same SAE family, per-frame, all layers, cross-model universality), but our contribution is orthogonal in substance: confound-controlled *emotion*-feature identification on a lexicon-controlled factorial corpus across six encoders, the decodability $\neq$ disentanglement-by-depth finding, and the depth-localized EMIS lexical shortcut—none of which AudioSAE, the ASR-SAE works [12, 13], or the non-SAE mechanistic SER work [11] provide. The combination is, to our knowledge, unscooped. A workshop or short-paper venue at an affective-computing or interpretability track is a realistic target, framing AudioSAE as concurrent method work and our affect-specific, confound-controlled analysis as the new contribution.

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